1. **Scenario:** You are developing a banking application that categorizes transactions based on the amount entered.
   Write logic to determine whether the amount is positive, negative, or zero.

   Read the amount number.
   if the number is greater than 0 print “positive”(Deposit)
   Else if the number is lesser than 0 print “negative”(withdrawal)
   Else ,print “number is zero” ”(No transaction)

2. **Scenario:** A digital locker requires users to enter a numerical passcode. As part of a security feature, the system checks the sum of the digits of the passcode.
   Write logic to compute the sum of the digits of a given number.

   Read the input number
   Convert the number into individual digits.
   Initialize a sum variable to 0.
   For each digit in the number, add it to the sum variable.
   Print the sum of the digits.

3. **Scenario:** A mobile payment app uses a simple checksum validation where reversing a transaction ID helps detect fraud.
   Write logic to take a number and return its reverse.

   Read the input transaction id
   convert the number into string
   reverses the string
   Convert it back to a number.
   Print the reversed number.

4.Scenario: In a secure login system, certain features are enabled only for users with prime-numbered user IDs.
Write logic to check if a given number is prime.

Read the input number
if the number is less than 2,print “Not prime”
Check divisibility of the number from 2 up to the square root of the number.
If it is divisible by any number in that range print “**not prime”**.
Otherwise print “**prime**”

5.Scenario: A scientist is working on permutations and needs to calculate the factorial of numbers frequently.
Write logic to find the factorial of a given number using recursion.

Read the input number.
If the number is 0 or 1, return 1.
Else, return the number multiplied by the factorial of (number - 1).
Print the result.

6.Scenario: A unique lottery system assigns ticket numbers where only Armstrong numbers win the jackpot.
Write logic to check whether a given number is an Armstrong number.

Read the input number.
Count the number of digits.
Initialize a sum variable to 0.
For each digit in the number:
Raise the digit to the power of the total number of digits.
Add the result to the sum variable.
If the sum is equal to the original number, print "Armstrong Number".
Else, print "Not an Armstrong Number".

- **7.Scenario:** A password manager needs to strengthen weak passwords by swapping the first and last characters of user-generated passwords.
  Write logic to perform this operation on a given string.

  Read the input string.
  If the string length is less than 2, print the string as is.
  Swap the first and last characters while keeping the middle part unchanged.
  Print the modified string.

8.Scenario: A low-level networking application requires decimal numbers to be converted into binary format before transmission.
Write logic to convert a given decimal number into its binary equivalent.

Read the decimal number as input
Initialize an empty string store for binary digits
Repeatedly divide the number by 2 and store the remainder.
Continue until the number becomes 0
Update the number by dividing it by 2.
Reverse the remainder binary number.
Print the binary equivalent.

9.Scenario: A text-processing tool helps summarize articles by identifying the most significant words.
Write logic to find the longest word in a sentence.

Read the sentence as input.
Split the sentence into words.
Initialize variables to keep track of the longest word and its length.
Iterate through the words, update the longest word if a longer one is found.
Print the longest word.

- **10.Scenario:** A plagiarism detection tool compares words from different documents and checks if they are anagrams (same characters but different order).
  Write logic to check whether two given strings are anagrams.

  Read the two string as input.
  Remove spaces and convert both strings to lowercase.
  Sort the characters of both strings.
  If sorted strings are equal, print "Anagram".
  Else, print "Not an Anagram".